



Parking Simulator API Documentation

Endpoint Overview

Base Path: `/api/v1`

This API provides a way to discover and retrieve details about Park Simulator components, along with a separate set of endpoints for sending commands to and controlling those components.

HTTP Method	Endpoint Path	Authentication Required
POST	<code>/api/v1/auth/login</code>	No
GET	<code>/api/v1/list-parking-spots</code>	Yes
GET	<code>/api/v1/list-barriers</code>	Yes
GET	<code>/api/v1/list-lights</code>	Yes
GET	<code>/api/v1/list-exhaust-fans</code>	Yes
GET	<code>/api/v1/list-alarms</code>	Yes
GET	<code>/api/v1/list-zones</code>	Yes
GET	<code>/api/v1/test</code>	Yes
POST	<code>/api/v1/barrier-gates/{name}/open</code>	Yes
POST	<code>/api/v1/barrier-gates/{name}/close</code>	Yes
POST	<code>/api/v1/barrier-gates/{name}/repair</code>	Yes
POST	<code>/api/v1/lights/{name}/on</code>	Yes
POST	<code>/api/v1/lights/{name}/off</code>	Yes
POST	<code>/api/v1/lights/group/{name}/on</code>	Yes
POST	<code>/api/v1/lights/group/{name}/off</code>	Yes
POST	<code>/api/v1/exhaust-fans/{name}/repair</code>	Yes
POST	<code>/api/v1/exhaust-fans/{name}/on</code>	Yes
POST	<code>/api/v1/exhaust-fans/{name}/off</code>	Yes
POST	<code>/api/v1/parking-spots/{name}/repair</code>	Yes
POST	<code>/api/v1/car/{name}/goto/{destination}</code>	Yes

POST	<code>/api/v1/car/{name}/charge?parkingCost=0&chargingCost=0</code>	Yes

1) Authentication:

POST `/api/v1/auth/login`

Authenticates user credentials and returns a JWT Bearer token for subsequent requests.

- **Authentication Required:** No
- **Request Body (JSON):**
 - `email` (string, required): User email address.
 - `password` (string, required): User password.
- **Return Type:**
 - `200 OK` :

```
{"token": "<JWT_TOKEN_STRING>"}
```
 - `401 Unauthorized` : Returned when authentication fails.

2) Component Listing & Diagnostics Endpoints

- These control APIs provide a way to discover the simulator's current level and state. They should be used only when discovery or synchronization is required. Periodic polling is not recommended, as each request carries a **simulated operational cost**. So please use it **ONLY once per level loading**. or after crash to do sync.

GET /api/v1/list-parking-spots

Returns a list of all parking spots and entry/exit spots along with their current status and detected vehicle plates.

- **Authentication Required:** Yes (**Bearer <token>**)
- **Parameters:** None
- **Return Type:** **200 OK**
- purpose can be **Park**, **EntrySpot**, **ExitSpot** , parkingForCarType can be: **Electric**, **Accessible**, **Any**, detected cars are list of cars detected in that spot, Zone parent is the name of Zone contains this parking spot, broken means: the spot is already broken, isUnderMaintenance means if there is a maintenance operation took place or not

```
[ { "name": "S150", "purpose": "Park", "parkingForCarType": "Any",  
  "zoneParent": "ZONE1", "detectedCars": [], "broken": false,  
  "isUnderMaintenance": false }, { "name": "S151", "purpose": "Park",  
  "parkingForCarType": "Electric", "zoneParent": "ZONE1", "detectedCars": [],  
  "broken": false, "isUnderMaintenance": false },
```

GET /api/v1/list-barriers

Returns a list of all barrier gates and their operational states.

- **Authentication Required:** Yes (**Bearer <token>**)
- **Parameters:** None
- **Return Type:** **200 OK**

- state can be { Open, Closed, Opening, Closing }

```
[ { "name": "gate0", "zoneParent": "", "broken": false,
  "isUnderMaintenance": false, "state": "Open" }, { "name": "gate1",
  "zoneParent": "ZONE1", "broken": false, "isUnderMaintenance": false,
  "state": "Closed" } ]
```

GET /api/v1/list-lights

Returns a list of all lighting fixtures and their state.

- **Authentication Required:** Yes (Bearer <token>)
- **Parameters:** None
- **Return Type:** 200 OK
- group is the group name used to control all lights in that group

```
[ { "name": "t_0", "group": "G1", "zoneParent": "ZONE1", "isOn": true }, {
  "name": "t_0", "group": "G1", "zoneParent": "ZONE1", "isOn": true }, ]
```

GET /api/v1/list-exhaust-fans

Returns a list of all exhaust ventilation fans and their status.

- **Authentication Required:** Yes (Bearer <token>)
- **Parameters:** None
- **Return Type:** 200 OK
- Sample response:

```
[ { "name": "fan0", "zoneParent": "", "broken": false,
  "isUnderMaintenance": true, "isOn": false } ]
```

GET /api/v1/list-alarms

Returns a list of all components currently flagged as broken and needing maintenance.

- **Authentication Required:** Yes (Bearer <token>)
- **Parameters:** None
- **Return Type:** 200 OK

```
[ { "name": "S150", "problem": "Require Maintenance" }, { "name": "S151",
  "problem": "Require Maintenance" }, { "name": "S152", "problem": "Require
  Maintenance" },
```

GET /api/v1/test

Triggers a test webhook event and returns the webhook endpoint confirmation.

- **Authentication Required:** Yes (Bearer <token>)
- **Parameters:** None
- **Return Type:** 200 OK

"Please wait for webhook call on http://your-webhook-url"

GET /api/v1/list-zones

Returns the current zone risk and gas concentration data.

- **Authentication Required:** Yes

- Parameters: None
- Return Type: 200 OK
- Response Example:

```
[ { "name": "ZONE1", "gasCarbonMonoxideLevel": 0, "risk": "Safe" }, {  
  "name": "ZONE2", "gasCarbonMonoxideLevel": 0, "risk": "Safe" } ]
```

- Field Meaning:
 - `name` : Zone name
 - `gasCarbonMonoxideLevel` : Carbon monoxide concentration in the zone
 - `risk` : Computed danger state, such as `"Low"` , `"Moderate"` , or `"High"`

3) Control Components

Barrier Gates Control Endpoints

POST /api/v1/barrier-gates/{name}/open

Sends a command to open the specified barrier gate.

- Authentication Required: Yes (`Bearer <token>`)
- Path Parameters:
 - `name` (string, required): The identifier name of the barrier gate.
- Return Type: `201 Created` (Empty body)

POST /api/v1/barrier-gates/{name}/close

Sends a command to close the specified barrier gate.

- **Authentication Required:** Yes (`Bearer <token>`)
 - **Path Parameters:**
 - `name` (string, required): The identifier name of the barrier gate.
 - **Return Type:** `201 Created` (Empty body)
-

POST /api/v1/barrier-gates/{name}/repair

Initiates repair on the specified barrier gate.

- **Authentication Required:** Yes (`Bearer <token>`)
 - **Path Parameters:**
 - `name` (string, required): The identifier name of the barrier gate.
 - **Return Type:** `201 Created` (Empty body)
-

Lighting Control Endpoints

POST /api/v1/lights/{name}/on

Turns on an individual light component.

- **Authentication Required:** Yes (`Bearer <token>`)
 - **Path Parameters:**
 - `name` (string, required): The identifier name of the light.
 - **Return Type:** `201 Created` (Empty body)
-

POST /api/v1/lights/{name}/off

Turns off an individual light component.

- **Authentication Required:** Yes (`Bearer <token>`)
 - **Path Parameters:**
 - `name` (string, required): The identifier name of the light.
 - **Return Type:** `201 Created` (Empty body)
-

`POST /api/v1/lights/group/{name}/on`

Turns on all lights associated with a specific group identifier.

- **Authentication Required:** Yes (`Bearer <token>`)
 - **Path Parameters:**
 - `name` (string, required): The group identifier name for the target lights.
 - **Return Type:** `201 Created` (Empty body)
-

`POST /api/v1/lights/group/{name}/off`

Turns off all lights associated with a specific group identifier.

- **Authentication Required:** Yes (`Bearer <token>`)
 - **Path Parameters:**
 - `name` (string, required): The group identifier name for the target lights.
 - **Return Type:** `201 Created` (Empty body)
-

Exhaust Fans Control Endpoints

`POST /api/v1/exhaust-fans/{name}/repair`

Initiates repair on an exhaust fan.

- **Authentication Required:** Yes (`Bearer <token>`)
- **Path Parameters:**
 - `name` (string, required): The identifier name of the exhaust fan.

- **Return Type:** `201 Created` (Empty body)
-

POST /api/v1/exhaust-fans/{name}/on

Turns on a specific exhaust fan.

- **Authentication Required:** Yes (`Bearer <token>`)
 - **Path Parameters:**
 - `name` (string, required): The identifier name of the exhaust fan.
 - **Return Type:** `201 Created` (Empty body)
-

POST /api/v1/exhaust-fans/{name}/off

Turns off a specific exhaust fan.

- **Authentication Required:** Yes (`Bearer <token>`)
 - **Path Parameters:**
 - `name` (string, required): The identifier name of the exhaust fan.
 - **Return Type:** `201 Created` (Empty body)
-

P Parking Spot Control Endpoints

POST /api/v1/parking-spots/{name}/repair

Initiates repair on a parking spot sensor or mechanism.

- **Authentication Required:** Yes (`Bearer <token>`)
 - **Path Parameters:**
 - `name` (string, required): The identifier name of the parking spot.
 - **Return Type:** `201 Created` (Empty body)
-

Car Control Endpoints

POST /api/v1/car/{name}/goto/{destination}

Moves a car to a parking destination or to an exit/escape route.

- **Authentication Required:** Yes
- **Path Parameters:**
 - **name** (**string** , required): The car identifier. the plate number with or without spaces
 - **destination** (**string** , required): The target destination:
 - A parking spot name
 - **"exit"** to leave via any exit (will ask for payment there)
 - **"leavepark"** to leave via any escape route
- **Return Type:** **201 Created**
 - Response body is empty.
- **Notes:**
 - If the target parking spot is already occupied a penalty may be registered.

Ask Car for payment

POST /api/v1/car/{name}/charge

Charges a car for parking and, if applicable, electricity usage.

- **Authentication Required:** Yes
- **Path Parameters:**
 - **name** (**string** , required): The car identifier, by plate or name

- **Query/Form Body Parameters:**
 - `parkingCost` (`float` , required): Parking charge amount